

PHASE ONE PROJECT

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Title:

Aviation Accident Analysis

Year Range: 1962–2023

OVERVIEW

Main objective:

Aims to deliver data-driven insights that would provide insights in efforts to improve air safety

Specific objectives:

- To show the most common causes of aviation accidents
- To show how accident rates vary by aircraft type and location
- To understand the role of weather in aviation accident

Data Source:

A csv file from National Transportation Safety Board (NTSB) with accident reports ranging 1962-2023

Business Understanding:

Purpose of this analysis is based on understanding aviation accident trends to improve safety and operations.

DATA UNDERSTANDING

Data Composition:

- Accident date, location, aircraft type, fatalities and contributing factors
- Missing values handled using imputation techniques.
- **Comments on the data**
 - As part of data cleaning, addressed duplicates and handled missing values
 - Derived insights from temporal trends.

DATA ANALYSIS

Some of the analysis I have managed to analyse and make visualizations included:

- ❖ Accidents trend over time
- ❖ Severity of accidents
- ❖ Phase of flights where most accidents take place
- ❖ Fatal and non fatal accidents by weather
- ❖ Number of accidents has influenced by weather
- ❖ Categorization of accidents based on purpose of flights
- ❖ The number of accidents for every make of aircrafts

Recommendation

- The makes recording highest number of accidents be assessed to rule out technical malfunction and so that safety measures on the crafts can be checked and improved
- Avoid flying in bad weather conditions
- The data should have included fatalities caused by pilots; maybe categorized in terms of pilot skill level, level of rest before the flight and number of trips in the same route. This would make the accident analysis more comprehensive

Next Step:

- Would be great to do a predictive model for accident risk assessments and monitoring
- Be able to get real time feedback and insights on the dashboard



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